## Principles of Vaccination

- The primary goal in vaccination is to provide protective immunity by inducing a memory response to an infectious microorganism using a non-toxic antigen preparation. It is important to produce immunity of the appropriate kind: antibody / or cellular immunity.
- Antibodies produced as a result of immunization are effective primarily against extracellular organisms and their products e.g., toxins. Passively administered antibodies have the same effect as induced antibodies.
- Cell-mediated immunity (T cells, macrophages) induced by vaccination is important particularly in preventing intracellular bacterial and viral infections and fungal infections.
- The ultimate goal of any immunization program is the eradication of the disease.
- This requires that the infection is limited only to humans, with no animal or environmental reservoir, and the absence of any subclinical or carrier state in humans.
- Achieving elimination requires a high level of herd immunity to prevent person to person spread.
- This requires considerable infrastructure support to ensure that all at-risk populations are targeted for immunization.
- This has been achieved for small pox, although we are close to the elimination of polio.

## Vaccine Types

- **Live attenuated vaccines:** These vaccines contain modified strains of a pathogen that have been weakened but are able to multiply within the body and remain antigenic enough to induce a strong immune response. Example: Oral Polio vaccine
- **Heterologous vaccine:** These are a group of live attenuated vaccines produced from the strains that are pathogenic in animals and not in humans. It is a vaccine that confers protective immunity against a pathogen that shares cross-reacting antigens with the microorganisms in the vaccine. Example: Cow pox virus that protects against small pox in humans.
- **Killed inactivated vaccines:** These groups of vaccine are produced either by killing or inactivating the bacteria or virus by chemical treatment or heat. Example:; Polio virus
- **Sub unit vaccine:** Uses the antigenic determinant / epitope (the very specific part of the microbe) is used to prepare the vaccine.
- **DNA Vaccines**: When the genes for microbe's antigens are introduced into the body some cells will take up the DNA. The DNA then instructs those cells to make the antigen molecules. The cells secrete the antigens adn display them on their surfaces. The body's own cells become vaccine generating factories.



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